
CMP6200/DIG6200
Individual Undergraduate Project (FYP)
Interim Progress Review
Sign Language Translation System

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Course: B.Sc. (Honors) Computer Science

Supervisor: Dr. Vi Chi Thanh

Reporting Period: 07/12/2025 – 10/01/2025

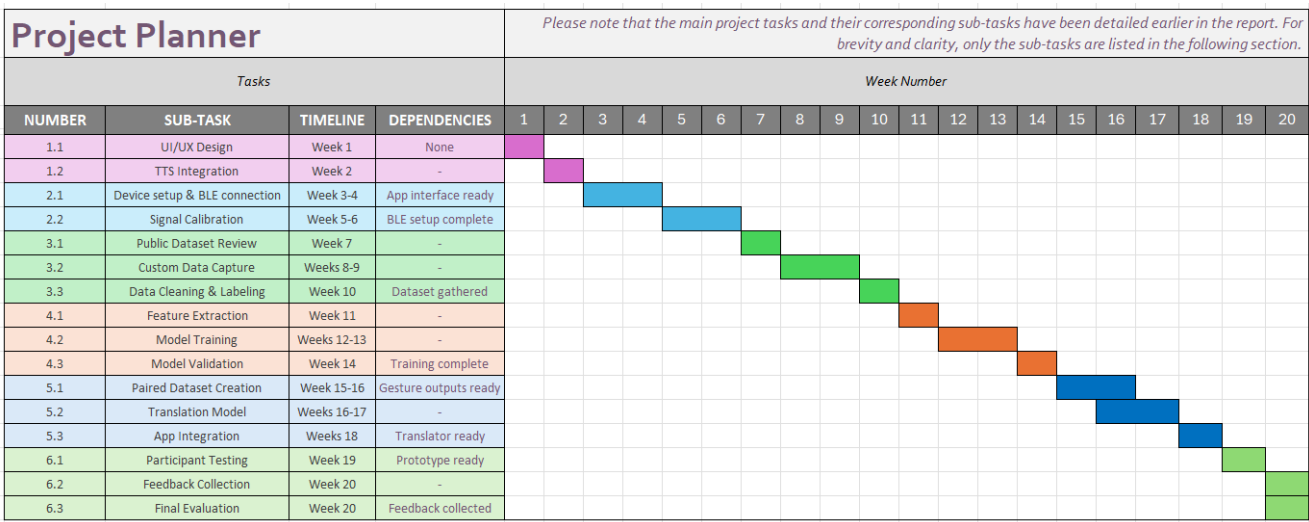
1.0 Log of Completed Activities & Milestones

This is a record of the activities done so far (milestones):

- **UI/UX Design:** Successfully completed high-fidelity wireframes using Figma, which encompass onboarding, gesture calibration and live translation screens.
- **Usability Testing:** Conducted testing with 5 participants (Hybrid: In Real Life and Remote). Achieved an average SUS score of 89.0 (Excellent).
- **Environment Setup:** Established the Android Studio project environment and incorporated essential dependencies (e.g., CameraX for gesture recognition and Bluetooth API for wristband pairing).
- **Ethics & Documentation:** Completed participant information sheets and consent forms pertinent to user research.

2.0 Reflection and Iteration

Attached below is the Gantt chart initially proposed:



2.1 Progress Against Original Timeline

The project has successfully transitioned from the conceptual and design phase into the implementation phase. While the recruitment and scheduling for usability testing took slightly longer than the initial Gantt chart estimated, the high quality of the data collected has streamlined the development process. By validating the UI/UX logic through a high-fidelity Figma prototype first, I have avoided costly design changes during the current Android Studio coding phase. All core milestones like design finalization, usability testing and environment setup have been achieved within the reporting period.

Current Project Status:

The project is currently approximately two to three weeks behind the initial Gantt chart schedule. However, this delay is the result of a strategic and deliberate pivot intended to secure the project's long-term viability. I made the executive decision to extend the UX Evaluation and Prototyping phase beyond its original deadline. My rationale was that the technical complexity of sign language translation requires a flawless user interface logic; committing to a codebase without absolute user validation would have introduced high risk. This extension was successful, as evidenced by the 89.0 SUS score (Excellent), which now serves as a stable foundation for all upcoming development.

Furthermore, additional time has been invested in strategic resource planning. Due to the current uncertainty regarding university funding for high-end sEMG/IMU sensors, I have dedicated significant time to exploring alternative hardware pathways, including 3D printing and DIY sensor integration, to ensure the project reaches completion regardless of external financial factors.

2.2 Challenges Encountered & Resolutions

Challenge Encountered	Detail of the Problem	Resolution / Action Taken
Prioritizing UX over Development Speed	There was a risk that rushing into Android Studio implementation would result in a "technically sound" app that was difficult for the Deaf community to use.	Resolution: I opted to extend the testing period with the Figma prototype to "stress-test" the calibration and translation logic. Achieving an 89.0 SUS score confirmed that the UX is optimized, which actually reduces the likelihood of time-consuming UI refactoring during the coding phase.
Hardware Funding & Procurement Risks	The original plan relied on external university funding for professional wearable sensors. If this funding is delayed or denied, the physical component of the project would be at a standstill.	Resolution: I have proactively begun studying Additive Manufacturing (3D Printing). I am researching how to design and print a custom ergonomic wristband housing. By exploring DIY sensor alternatives and 3D printing, I am creating a "Plan B" that allows the project to proceed as a custom-built hardware-software solution if funding falls through.
Architectural Decision-Making	I needed to determine the best approach for the app's backend to ensure it could handle different types of	Resolution: I took extra time to study and implement a Modular/Sensor-Agnostic Architecture in Android Studio. This research ensures that the

	hardware (professional vs. DIY sensors).	code I write now can easily be "swapped" to work with whichever hardware solution we ultimately finalize, providing maximum technical flexibility.
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3.0 Action Plan: Goals for Next Period

- **3D Modeling & Prototyping:** Complete the first iteration of the model and perform a test print to check for fit and sensor placement.
- **Android Mock-Service:** Develop a robust "Mock Data Service" in Android Studio to test the translation engine logic while the hardware path is being finalized.
- **Funding Submission:** Prepare and present a detailed "Alternative Manufacturing Report" to the supervisor to demonstrate the cost-savings of a 3D-printed hardware approach.
- **Backend Integration:** Implement the local **SQLite/Room Database** to handle translation history and user profiles, ensuring the app is functional in an offline environment.

Key objectives to be met by Sunday, 18/01/2026:

By the next meeting, I hope to be able to

- Present a semi-completed Application on Android Studio as per the high-fidelity prototypes attached below:

The image displays a four-screen onboarding flow for a mobile application, presented as a wireframe with a light gray background and black text.

- Screen 1: SignSpeak**
 - Header: A blue circle containing a white hand icon.
 - Title: **SignSpeak**
 - Description: Translate sign language into text and speech in real time using your wearable device.
 - Features (in a light blue box):
 - ✓ Real-time translation
 - ✓ Speech output
 - ✓ Learning mode
 - ✓ Privacy-first design
 - Primary Button: **Get Started** (blue)
 - Footer: By continuing, you agree to our Terms & Privacy Policy
- Screen 2: Create Account**
 - Header: **Create Account**
 - Text: Sign up to get started
 - Buttons: **Continue with Google** and **Continue with Apple** (both white with gray borders)
 - Text: Or continue with email
 - Form: **Email Address** (input field with placeholder: your.email@example.com)
 - Form: **Password** (input field with placeholder: Enter your password)
 - Text: Minimum 8 characters
 - Primary Button: **Sign Up** (blue)
 - Text: [Already have an account? Sign in](#)
- Screen 3: Welcome Back**
 - Header: **Welcome Back**
 - Text: Sign in to continue
 - Buttons: **Continue with Google** and **Continue with Apple** (both white with gray borders)
 - Text: Or continue with email
 - Form: **Email Address** (input field with placeholder: your.email@example.com)
 - Form: **Password** (input field with placeholder: Enter your password)
 - Text: Minimum 8 characters
 - Primary Button: **Sign In** (blue)
 - Text: [Don't have an account? Sign up](#)
- Screen 4: Tell Us About Yourself**
 - Header: **Tell Us About Yourself**
 - Text: This helps us personalize your experience and enable the right features.
 - Info Box: You can change this later in Settings. Your choice affects which features are shown.
 - Options (each in a white box with a gray border):
 - Deaf User** (icon: crossed-out speaker): I use sign language and want to translate it to text and speech for others to understand.
 - Mute User** (icon: microphone with slash): I cannot speak and want to use sign language or gestures to communicate with others.
 - Hearing User** (icon: ear): I can hear and speak, and want to communicate with deaf or mute individuals.
 - Learner** (icon: graduation cap): I want to learn sign language and practice with feedback and tutorials.

← Back

This helps us personalize your experience and enable the right features.

You can change this later in Settings. Your choice affects which features are shown.



Deaf User

I use sign language and want to translate it to text and speech for others to understand.



Mute User

I cannot speak and want to use sign language or gestures to communicate with others.



Hearing User

I can hear and speak, and want to communicate with deaf or mute individuals.



Learner

I want to learn sign language and practice with feedback and tutorials.

Continue

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Pair Your Wearable

Connect your sEMG/IMU wristband to enable sign language translation.

Before you start:

1. Turn on your wearable device

2. Make sure Bluetooth is enabled on your phone

3. Keep the device within 3 feet of your phone

Scan for Devices

Skip for now (limited functionality)

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Scan for Devices

Available Devices:



SignSpeak Band - #A1B2C3



SignSpeak Band - #D4E5F6

Skip for now (limited functionality)

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Device Calibration

Follow the instructions to calibrate your wearable for accurate gesture recognition.

You'll perform these gestures:

1

Rest Position

Keep your hand relaxed at your side

2

Fist Gesture

Make a fist with your hand

3

Open Palm

Open your hand with palm facing forward

4

Point Gesture

Point your index finger upward

5

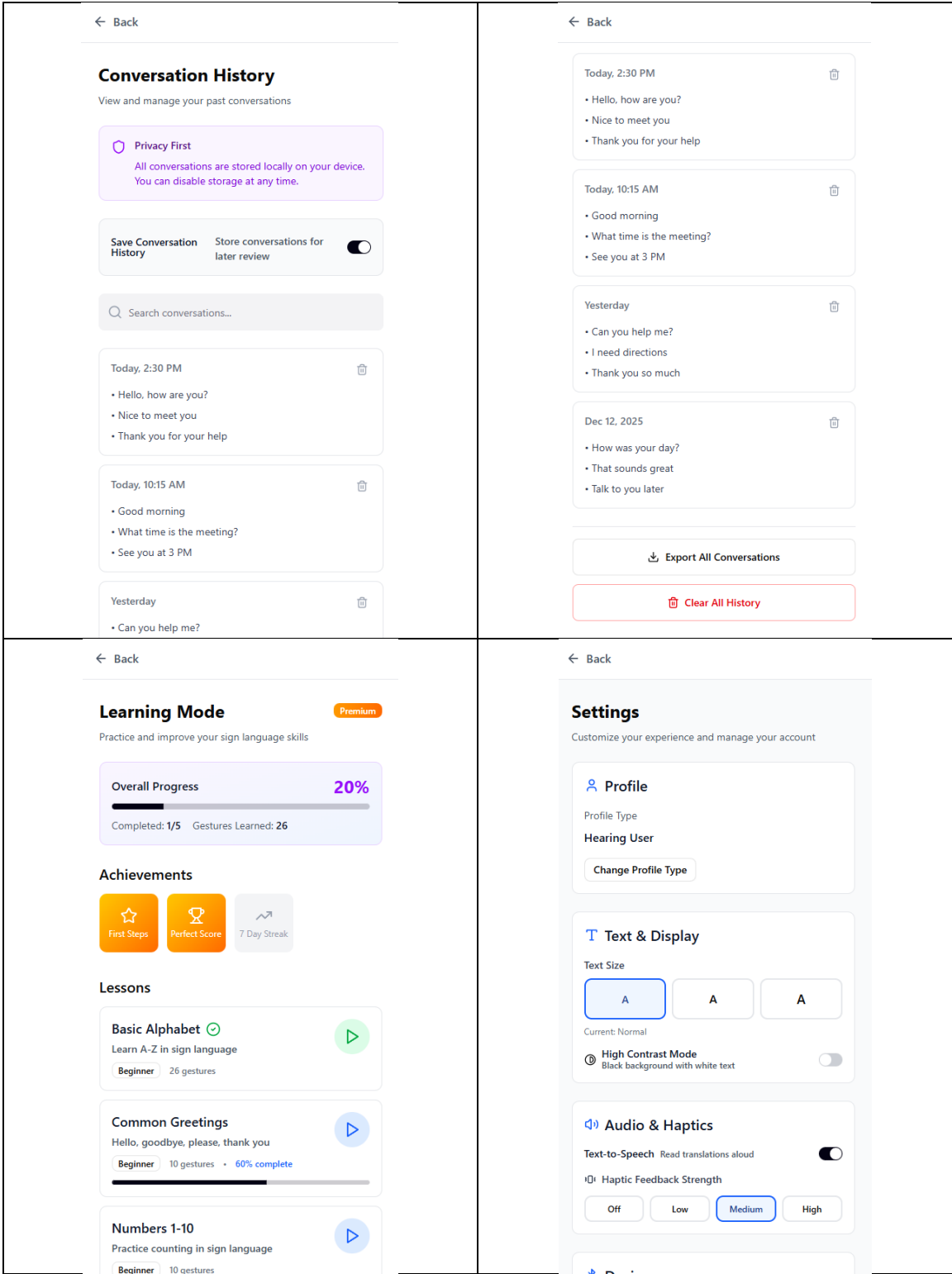
Wave Motion

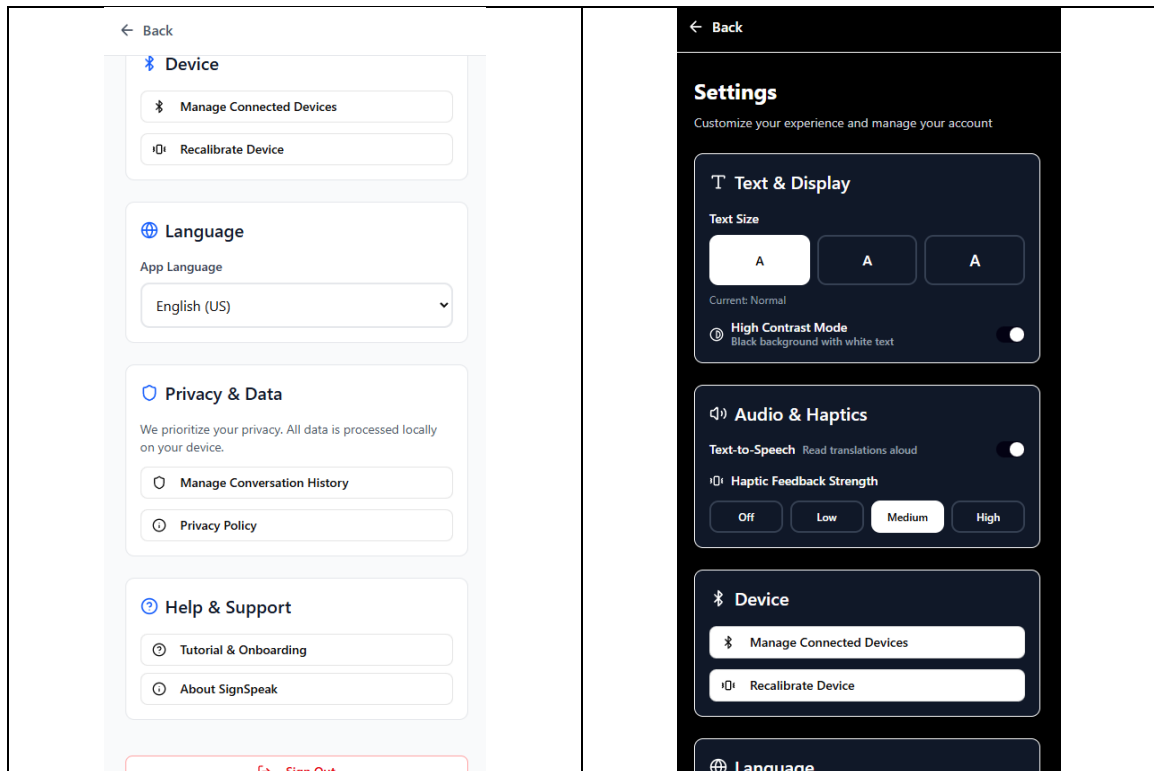
Wave your hand side to side

Tip: Hold each gesture steady for 3 seconds. The device will vibrate when each step is complete.

Start Calibration

← Back  Device Calibration Follow the instructions to calibrate your wearable for accurate gesture recognition. Step 2 of 5 1 Fist Gesture Make a fist with your hand Hold the gesture steady...	Live Translation Connected Signal: 94%  Start signing to see translation here...  TTS On  Speech  History  Learn  Settings
← Back Speech to Text Hearing users can speak, and the text will be displayed for deaf/mute users to read.  Tap the microphone button to start recording speech. How it works: Tap and hold the microphone to speak. Your speech will be converted to text in real-time. Release to stop. Microphone access is required for speech recognition	← Back Speech to Text Hearing users can speak, and the text will be displayed for deaf/mute users to read.  Recording 0:05  Hello, nice to meet you. How it works: Tap and hold the microphone to speak. Your speech will be converted to text in real-time. Release to stop. Microphone access is required for speech recognition





- Provide a rough idea of the hardware required for 3D-printing and building of the device in the likely event that the university funding path doesn't work out.
- Research/Study into the Speech-to-text and Text-to-speech engine required by this project.

4.0 Other Considerations

1. As outlined in the initial Project Timeline (Interim Report), the project was initially designed to adhere to a sequential flow. However, given the critical importance of the hardware-software integration, I have shifted to a Parallel Prototyping Methodology. This adjustment was essential to guarantee that the software development particularly the gesture recognition engine could advance despite the existing uncertainty surrounding sensor funding.
2. In addition to the change in approach, another major change is the potential move from commercial hardware to custom-printed hardware as this may give more design flexibility and may reduce the current cost of acquiring an EMG+IMU hardware like that of the Myo armband.

5.0 Questions for Supervisor

Below are my questions:

1. Since the university funding for professional sEMG sensors is currently unconfirmed, I am pivoting toward a 3D-printed enclosure. Given the wearable nature of the device, do you recommend I focus on TPU (Thermoplastic

Polyurethane) for its flexibility and skin-comfort, or should I stick to PLA/PETG for the initial structural prototype to ensure the electronic components are securely housed?

2. During my research into the Text-to-Speech (TTS) and Speech-to-Text (STT) components, I found that cloud-based APIs offer higher accuracy but introduce latency and internet dependency. For an Honors-level project, would you value the reliability of an offline-first approach (using Google ML Kit/Local TTS) more than the higher vocabulary range of a cloud-based service?
3. If I proceed with the 3D-printed/DIY hardware path, are there additional internal safety or ethical checks I need to complete through the university to ensure the custom electronic housing is cleared for further user testing sessions?